

Transformer Expressivity

English seminar study notes

These notes are an English companion to the seminar handout. The emphasis is on mathematical derivations that can be followed line by line, on separating theorem statements from interpretation, and on identifying the assumptions under which each claim is valid.

Primary readings

Yun et al., **Are Transformers Universal Approximators of Sequence-to-Sequence Functions?**, ICLR 2020.

Bhojanapalli et al., **Low-Rank Bottleneck in Multi-head Attention Models**, ICML 2020.

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1

Notation and the three levels of expressivity

1.1 Matrix conventions

Let a length- n sequence with token dimension d be represented as $X = [x_1, \dots, x_n] \in \mathbb{R}^{d \times n}$, so columns are tokens. A single attention head forms

$$Q = W_Q X, \quad K = W_K X, \quad V = W_V X,$$

and outputs, schematically,

$$\text{Attn}(X) = V \text{softmax}(K^\top Q),$$

where the softmax is column-wise under this convention. Other papers transpose every matrix; the mathematical content is unchanged if one is consistent.

1.2 Three questions that should not be mixed

1. **Block-level representation:** which attention matrices can one head produce?
2. **Network-level approximation:** which continuous sequence-to-sequence functions can a deep Transformer approximate?
3. **Efficient representation:** how much depth, width, and precision are required?

Universal approximation answers the second question, not automatically the first or third.

How to say it in the presentation A Transformer can be universal even if an individual attention head has a rank bottleneck. Universality is a statement about the composition of many blocks, not about unrestricted one-layer attention matrices.

2

Permutation symmetry of self-attention

2.1 Self-attention without positional information

Let $P \in \mathbb{R}^{n \times n}$ be a permutation matrix. Reordering the tokens gives XP . Then

$$Q(XP) = W_Q XP = Q(X)P,$$

and similarly $K(XP) = K(X)P$, $V(XP) = V(X)P$. Hence

$$K(XP)^\top Q(XP) = P^\top K(X)^\top Q(X)P.$$

Because column-wise softmax commutes with simultaneous row/column permutation,

$$\text{softmax}(P^\top AP) = P^\top \text{softmax}(A)P.$$

Therefore

$$\begin{aligned} \text{Attn}(XP) &= V(X)P P^\top \text{softmax}(K^\top Q)P \\ &= \text{Attn}(X)P. \end{aligned}$$

A token-wise feed-forward network also commutes with P , so the whole Transformer without positional information is permutation equivariant:

$$\boxed{T(XP) = T(X)P.}$$

2.2 What the symmetry excludes

If a target function F distinguishes absolute token positions, then $F(XP) \neq F(X)P$ for some P . Such a function cannot be uniformly represented by a position-free Transformer, no matter how it is trained.

How to say it in the presentation Before discussing universal approximation, identify the symmetry class. Without positional information, the correct target class is permutation-equivariant sequence-to-sequence functions.

3

Universal approximation: what the theorem actually says

3.1 L^p approximation on a compact domain

Let $K \subset \mathbb{R}^{d \times n}$ be compact and let $F : K \rightarrow \mathbb{R}^{d \times n}$ be continuous. A typical approximation statement uses

$$d_p(F, G) = \left(\int_K \|F(X) - G(X)\|_p^p dX \right)^{1/p}.$$

Universal approximation means: for every $\varepsilon > 0$, there exists a Transformer G in the specified architectural class such that $d_p(F, G) < \varepsilon$.

3.2 Yun et al.: fixed-width universality

A striking result of Yun et al. is that a Transformer architecture with a constant number of heads and constant head/feed-forward width can approximate every continuous permutation-equivariant sequence-to-sequence function on a compact domain. In their notation, the construction can be placed in a class often summarized as

$$\mathcal{T}_{2,1,4},$$

corresponding to two heads, scalar head size, and small feed-forward width, while depth is allowed to grow.

Caution “Fixed width” does not mean “constant total complexity.” The proof can require very large depth as the desired accuracy and input dimension change.

3.3 Why the result is plausible

The architecture need not store an arbitrary high-dimensional function in a single layer. Depth allows it to build a discrete contextual code, route information conditionally, and then decode the desired output.

How to say it in the presentation The surprising part of the theorem is not that a large Transformer is expressive. It is that width can remain bounded while depth carries the burden of universality.

4

The proof idea: contextual mappings

4.1 Why token-wise MLPs are insufficient

A token-wise map g has the form

$$G(X) = [g(x_1), \dots, g(x_n)].$$

It cannot distinguish the same token value appearing in different sequence contexts. Attention is needed to make a token representation depend on the rest of the sequence.

4.2 Contextual IDs

The proof constructs an intermediate map in which each token obtains a representation that uniquely identifies both its own quantized value and the relevant global context. Once these contextual IDs are distinct, a token-wise network can map them to the desired output values.

4.3 Three proof stages

The high-level proof is:

1. **Quantize.** Approximate the continuous target by a piecewise-constant function on a fine grid.
2. **Contextualize and route.** Use an idealized Transformer with hard selections to assign distinguishable contextual codes and implement the grid function.
3. **Approximate the idealized operations.** Replace hardmax/piecewise gadgets by ordinary softmax and ReLU blocks with sufficiently sharp parameters.

Continuity and compactness control the approximation error introduced by the grid and by replacing idealized operations.

Supplementary derivation The logical structure is the same as many classical UAT proofs: first reduce an arbitrary continuous function to a finite combinatorial object, then prove that the architecture can realize that object, then approximate the idealized components by the actual activation family.

How to say it in the presentation The core role of attention in the proof is to create context-dependent identifiers. Once a token knows “who it is in this sequence,” a small feed-forward network can decode the desired output.

5

Positional encoding as symmetry breaking

5.1 Adding absolute position

Let $E = [e_1, \dots, e_n]$ be a positional code and feed $X + E$ into the Transformer. A token permutation changes X but not the association between e_j and position j , so the permutation-equivariance constraint is broken.

5.2 Universality for arbitrary fixed-length sequence maps

Yun et al. show that with trainable positional encodings, the same basic architecture can approximate arbitrary continuous fixed-length sequence-to-sequence functions, not only permutation-equivariant ones.

5.3 The exact scope

The theorem is for a fixed sequence length n and compact input domain. It does not imply length generalization to $n' > n$, nor does it state that a learned positional encoding extrapolates algorithmically.

How to say it in the presentation Positional encoding is mathematically a symmetry-breaking device. It enlarges the target function class from permutation-equivariant maps to arbitrary continuous maps on a fixed-length sequence space.

6

The expressivity of a single attention matrix

6.1 The softmax inverse problem

Suppose we want a single head to produce a positive stochastic matrix A . Before softmax, the score matrix has the form

$$S = K^\top Q = X^\top W_K^\top W_Q X.$$

If the projected key/query dimension is d_k , then

$$\text{rank}(S) \leq d_k.$$

Softmax is nonlinear, but this low-dimensional score factorization still constrains which attention patterns can be produced.

6.2 Bhojanapalli et al.: sufficient dimension

Under suitable full-rank assumptions on the input representation and sufficiently large key/-query dimension, one can solve for projection matrices whose scores realize a desired positive stochastic attention pattern after softmax. The construction effectively works by taking a logarithm of the desired probabilities up to column-wise additive constants and factorizing the resulting score matrix.

6.3 Low-rank bottleneck

When d_k is too small relative to sequence length, there exist attention matrices that cannot be represented by a single head on a fixed input. The obstruction comes from the low-rank score factorization.

6.4 Why this does not contradict UAT

Deep Transformers can use multiple heads, intermediate nonlinear transformations, and many layers. A restriction on one attention matrix therefore does not imply a corresponding restriction on the function class of the whole deep network.

How to say it in the presentation There are two different notions of expressivity: the set of attention matrices one layer can instantiate, and the set of functions a deep Transformer can approximate. The first can be low-rank constrained while the second is universal.

7

Depth, width, and efficiency

7.1 Small width can hide enormous depth

A universal approximation theorem is existential. If the construction needs a grid with resolution δ in a dn -dimensional domain, the number of grid cells can scale like

$$O(\delta^{-dn}),$$

which exhibits the usual curse of dimensionality. A fixed-width network may therefore require impractically large depth.

7.2 Depth as sequential computation

With width fixed, depth provides repeated opportunities to compare, route, and rewrite token representations. In the UAT proof, this sequential resource is what compensates for small head and MLP widths.

7.3 What the theorem does not imply

Universal approximation does not imply:

- that SGD finds the constructive solution;
- polynomial sample complexity;
- efficient depth or parameter count;
- length extrapolation;
- computational tractability of inference;
- interpretability of learned attention weights.

How to say it in the presentation Expressivity is a feasibility statement. Learnability, efficiency, and generalization are separate questions, and later weeks study exactly those gaps.

8

Presentation checklist

8.1 Eight formulas to keep

$$\begin{aligned}
 Q &= W_Q X, \quad K = W_K X, \quad V = W_V X, \\
 T(XP) &= T(X)P \quad (\text{no positional information}), \\
 d_p(F, G) &= \left(\int_K \|F(X) - G(X)\|_p^p dX \right)^{1/p}, \\
 \mathcal{T}_{2,1,4} &\quad (\text{fixed-width UAT construction}), \\
 X &\mapsto X + E \quad \text{breaks permutation symmetry}, \\
 S &= K^\top Q, \\
 \text{rank}(S) &\leq d_k, \\
 &\text{single-head matrix expressivity} \neq \text{deep-network universality}.
 \end{aligned}$$

8.2 Common mistakes

- Omitting the fixed-length and compact-domain assumptions.
- Saying that positional encoding is required for permutation-equivariant universality.
- Treating fixed width as an efficiency theorem.
- Using the rank of $K^\top Q$ as if it directly bounded the rank of the post-softmax matrix.
- Inferring SGD learnability from an existential construction.

How to say it in the presentation Thirty-second conclusion: self-attention without positional information is permutation equivariant. Despite strong one-layer structural constraints, sufficiently deep fixed-width Transformers are universal for continuous equivariant maps; positional encoding breaks the symmetry and extends the result to arbitrary continuous fixed-length sequence maps.

Bibliography

- Yun, C. et al. *Are Transformers Universal Approximators of Sequence-to-Sequence Functions?* ICLR, 2020.
- Bhojanapalli, S. et al. *Low-Rank Bottleneck in Multi-head Attention Models.* ICML, 2020.